

Practical No. 9

Projects using MySQL, Node.js and Express.js, React.js

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Prn – 245109074

Batch – S5

Perform following problem statements using Node.js

Problem Statement 3: Simple Blog Application

- Create a simple blog platform where users can sign up, log in, and write, edit, or delete their blog posts. Other users can view the blog posts but not edit or delete them.
- User authentication (JWT).
- Blog post CRUD operations.
- Display all blog posts on the homepage, with individual post details accessible through a unique URL.
- Add comment functionality where users can comment on blog posts.

Code Link (Github) - <https://github.com/Atharv-K-979/WCE-Labs-Material/tree/main/Web-Dev-Design-Lab/Assignment9/BlogApplication>

Deployed Website (Live) - <https://wce-labs-material.onrender.com>

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Pulling Git Out of the Black Box

Most developers use Git daily without ever questioning what's happening beneath git add, git commit, or git push. I was one of them—until I decided to build a Git-inspired distributed version control system from scratch.

This project isn't a Git clone. It's an attempt to understand the engineering principles behind version control: state tracking, immutability, staging, and remote synchronization — by implementing them myself using a custom CLI, a REST backend, and cloud storage.

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```
graph LR
    subgraph Left
        SA[STAGING AREA]
        CD[COMMITTS DIRECTORY]
    end
    subgraph Backend [BACKEND SERVICES]
        EA[EXPRESS API]
        EA --> E[events]
        EA --> M[metadata]
        EA --> B[blobs]
        E -.-> SI[SOCKET.IO]
    end
    subgraph Cloud [CLOUD INFRASTRUCTURE]
        M --> MD[(MONGODB)]
        B --> MD
        MD --> AS3[(AWS S3)]
    end
    SA -- status --> EA
    CD -- status --> EA
```

atharv • less than a minute ago



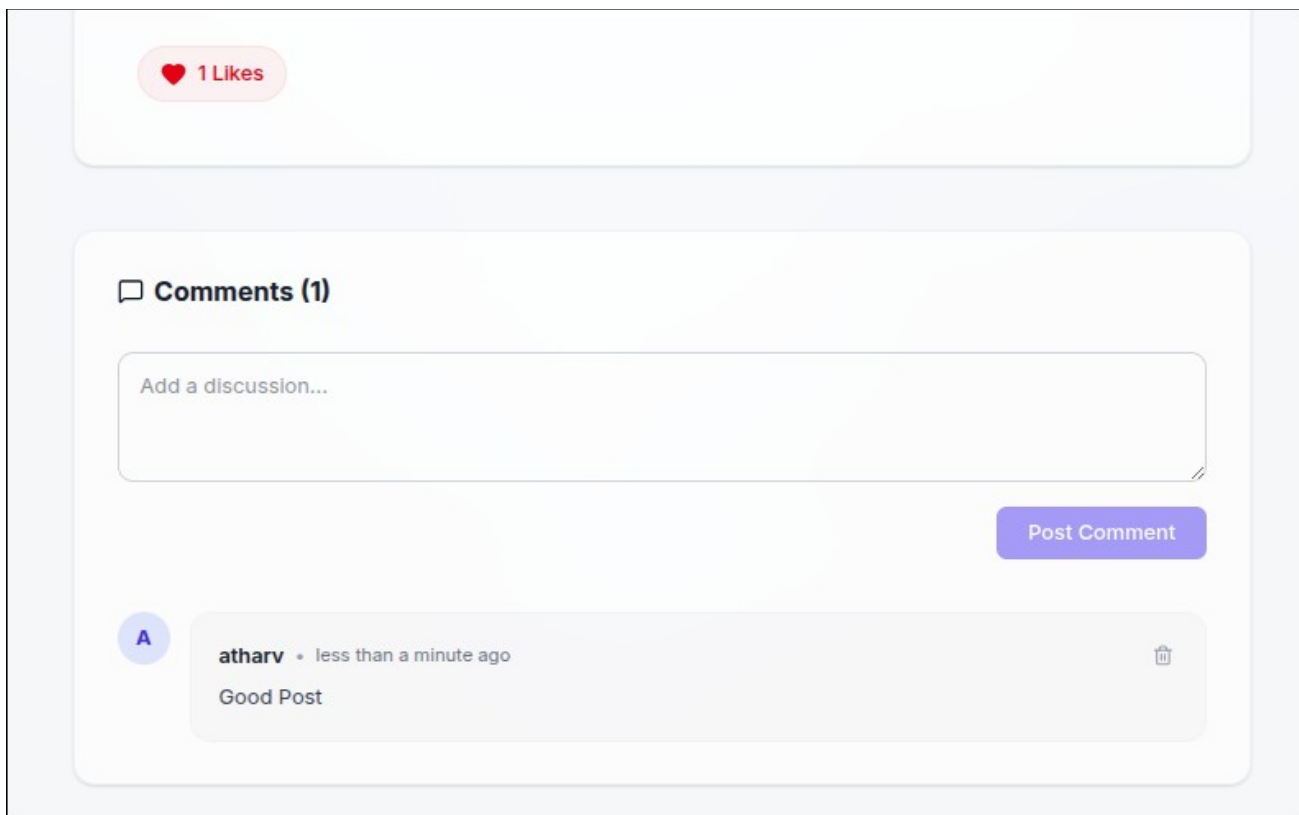
Building a Distributed Version Control System from Scratch

Pulling Git Out of the Black Box
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This project isn't a Git clone. It's an attempt to understand the engineering principles behind version control: state tracking, immutability, staging, and remote synchronization — by implementing them myself using a custom CLI, a REST backend, and cloud storage.

The result is a working VCS with a CLI and a GitHub-style web dashboard, built using the MERN stack and AWS S3.

High-Level Architecture



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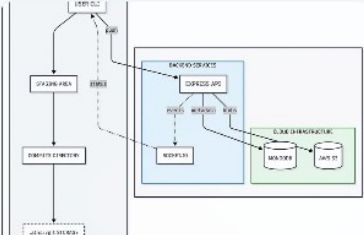
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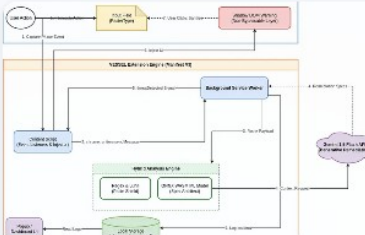
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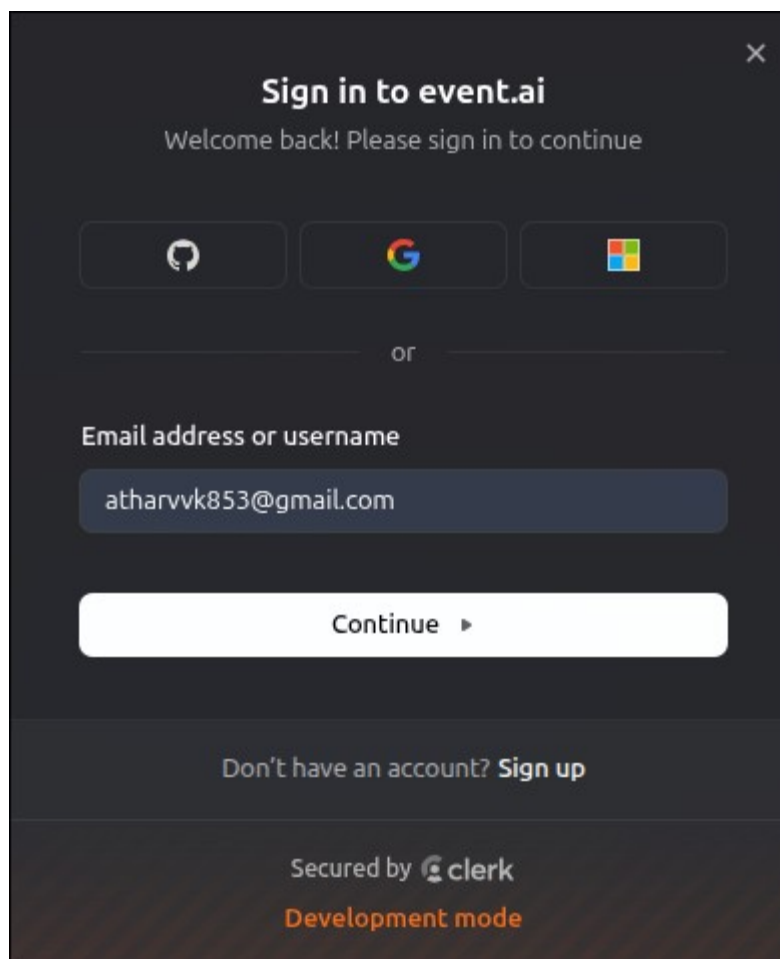


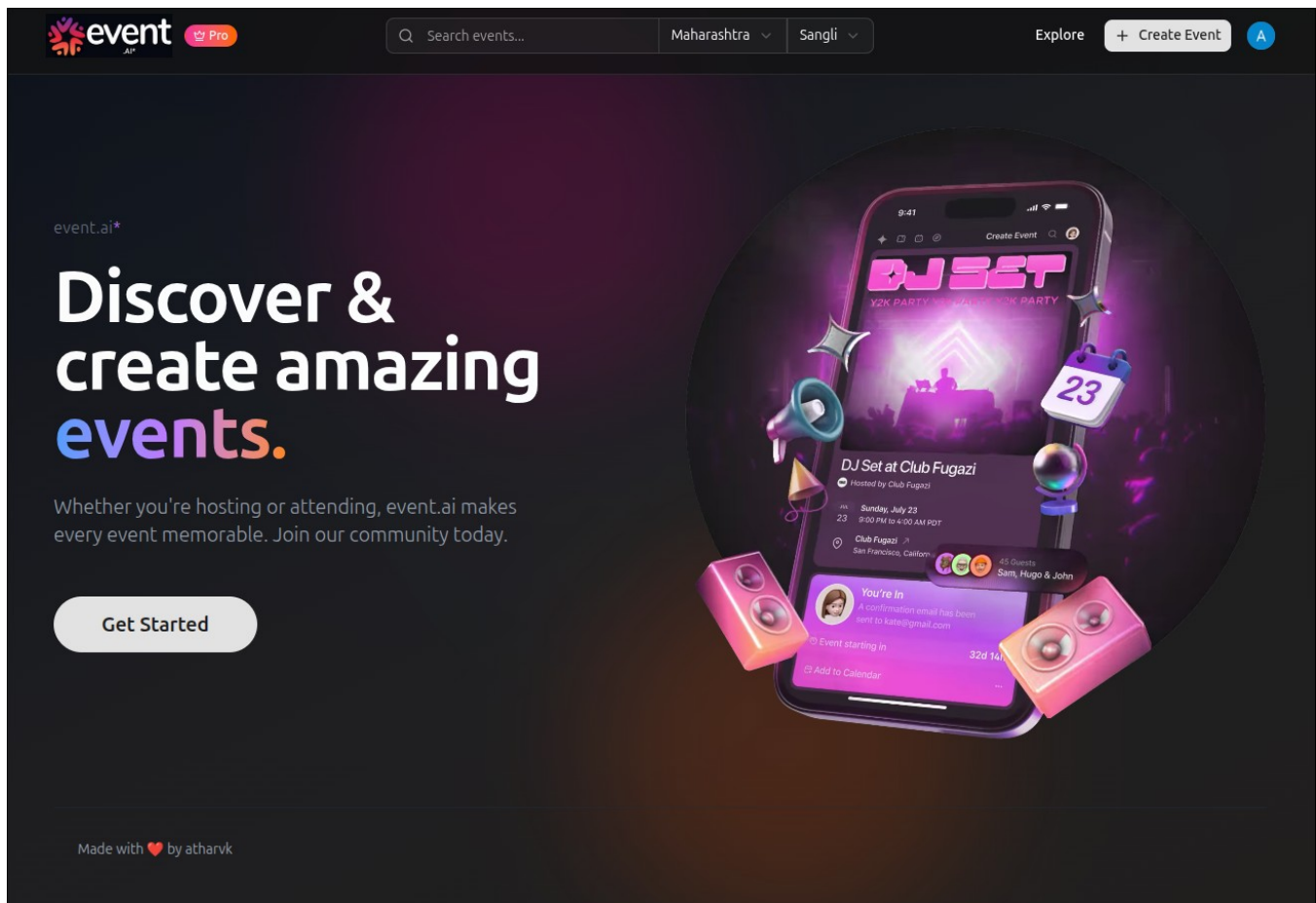
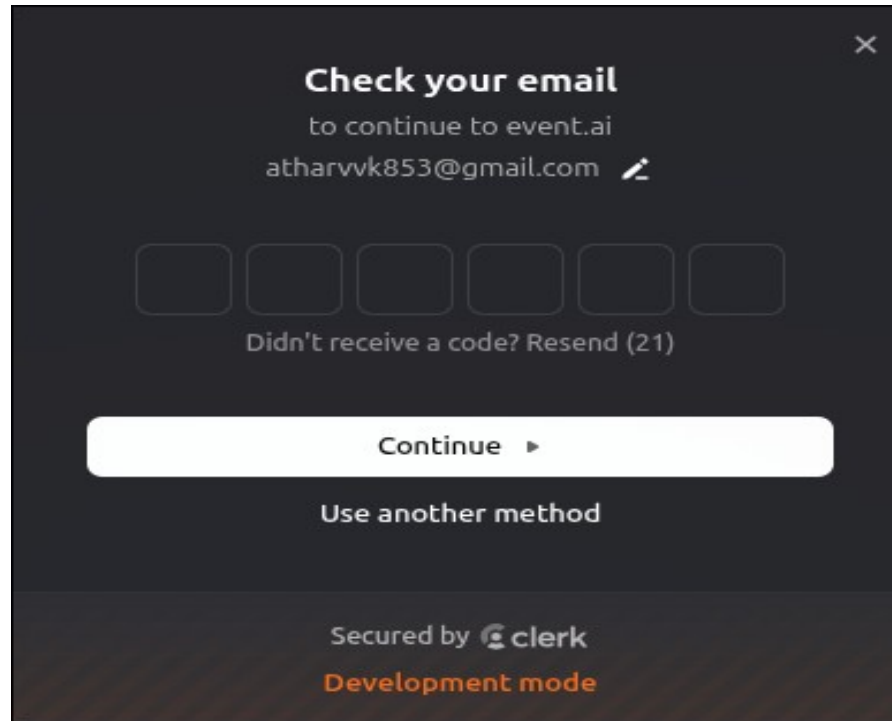
Problem Statement 5: Event Planner Application

- Create an event planner app where users can organize events, including event details such as event name, date, location, and description.
- User authentication and role-based access (e.g., admin and user).
- CRUD operations for events (only admins can create, edit, or delete events).
- Display event details for all users.
- Implement event registration functionality, where users can register for events.
- Add email notifications when users register for an event.

Code Link (Github) - <https://github.com/Atharv-K-979/event.ai.git>
Deployed Website (Live) – event-ai-ak.vercel.app

Screenshots -



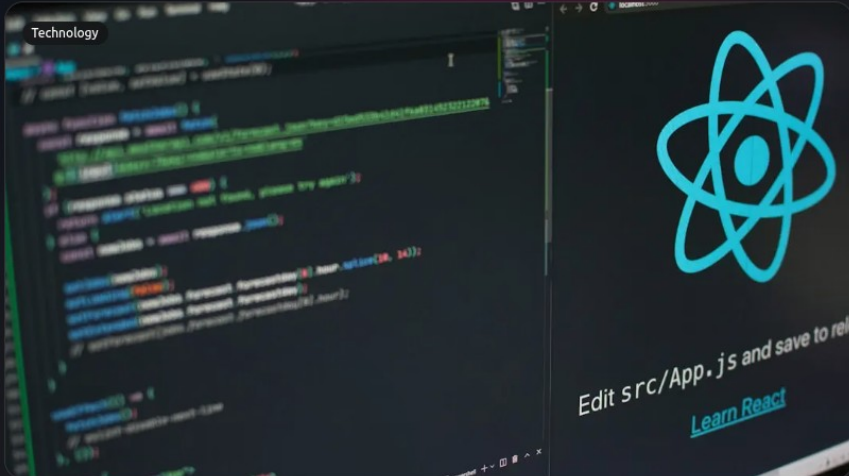




Search events...

MaharashtraSangli

Explore+ Create Event



Technology

Price\$250

Spots remaining54 / 75

Get Tickets

Share this event

React & Next.js: Building the Future of Web Applications

February 5th, 2026 10:05 AM Gurgaon, Haryana By Ramesh

About this event

Explore the latest advancements in React and Next.js, and discover how these technologies are shaping modern web development. This event provides insights into server-side rendering, performance optimization, and the future roadmap for building scalable and efficient web applications. Attendees will gain practical knowledge and explore best practices for leveraging these powerful tools.

The screenshot displays the 'event AI Pro' website interface for creating an event. The top navigation bar includes the logo, a search bar, location filters for 'Maharashtra' and 'Sangli', and buttons for 'Explore', 'Create Event', and a user profile icon. The main section is titled 'Create Event' and features a large placeholder for a cover image with the text 'Click to add cover image'. To the left of the form are 'Theme Color' options represented by colored circles. The form itself contains several input fields: 'Event Name', 'Start' and 'End' date pickers, a 'Category' dropdown, a 'Location' dropdown, a 'Venue' dropdown, and a 'Full address / street / building (optional)' text area. Below these are a 'Description' text area, 'Tickets' options (Free and Paid), and a 'Capacity' input field. An 'AI Event Assistant' modal is open, prompting the user to 'Describe your event idea, and let AI fill in the details.' It contains a text input with the example text 'I want to take Web Development Bootcamp on 28 april.' and a 'Generate Details' button. At the bottom right of the form is a large 'Create Event' button.

event AI Pro

Search events...

Maharashtra Sangli

Explore + Create Event

Create Event

Create with AI

Click to add cover image

Theme Color

Event Name

Start End

Pick date -- -- Pick date -- --

Category

AI Event Assistant

Describe your event idea, and let AI fill in the details.

I want to take Web Development Bootcamp on 28 april.

Generate Details

Location

Venue

Full address / street / building (optional)

Description

Tell people about your event...

Tickets

Free Paid

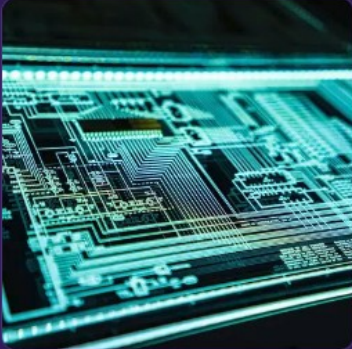
Capacity

50

Create Event

Create Event

Create with AI



Theme Color

AI Generated: I want to take Web Development Bootcamp on 28 april 2026

Start

April 28th, 2026

10:30 AM

End

April 28th, 2026

05:01 PM

Category

Technology

Location

Maharashtra

Sangli

Venue Details

Venue link (Google Maps Link)

Walchand College of Engineering Sangli

Description

This is an auto-generated description for "I want to take Web Development Bootcamp on 28 april 2026". It includes detailed information about the schedule, speakers, and venue highlights to attract attendees. Join us for an unforgettable experience!

Tickets

Free

Paid

99

Capacity

300

Create Event

